

Federico Giacardi

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Education

Master's Degree in Computer Science

Sep 2023 – Apr 2026

University of Turin | Final Mark: 110/110 cum laude with mention | GPA: 3.90/4

Relevant coursework: Machine Learning, Deep Learning, Computer Vision and Signal Processing, Natural Language Processing, Statistics and Simulation, Ethics and Fairness in AI, Numerical Computing, Parallel and Accelerated Computing, Combinatorial Optimization.

- Selected for the UniTO competitive programming team (ItaCPC 2023, **SWERC 2024**).
- Teaching assistant for “Computer Architectures”, “Programming I/II”, and “Operations Research” courses.
- Selected for **Neurohackademy 2026** summer school

Bachelor's Degree in Computer Science

Sep 2020 – Nov 2023

University of Turin | Final Mark: 110/110 cum laude | GPA: 3.78/4

Thesis: “Efficient Heuristics for Team Formation in Aggregate Computing”

Supervisor: Prof. Giorgio Audrito

Designed and implemented heuristic algorithms for automatic team formation among distributed autonomous devices using the Aggregate Computing paradigm.

- Selected for the UniTO competitive programming team (ItaCPC 2022).
- Teaching assistant for “Operating Systems” courses.

Research Experience

Master's Thesis

Oct 2025 – Apr 2026

INRIA Saclay (MIND team) and University of Turin

On-site at INRIA Saclay from January to March 2026, funded by Erasmus+ Traineeship and a UniTO Department of Computer Science international mobility scholarships

Supervisors: Carlo Alberto Barbano (INRIA), Demian Wassermann (INRIA), Marco Grangetto (UniTO)

Title: Predicting Cognition from Functional MRI with Machine Learning: Methodological and Data Challenges

- Built a reproducible benchmarking pipeline for in-domain and out-of-domain cognitive scores prediction from resting-state functional connectivity, replicating and extending the works of Ooi et al. (2022) and He et al. (2020) on HCP-YA and CamCAN.
- Designed and validated a domain adaptation method based on Optimal Transport in the log-Cholesky space, exploiting the SPD geometry of functional connectivity matrices.
- Implemented and analysed a gradient-reversal-based adversarial debiasing approach, showing its training instability and inverse-encoding failure mode.
- Confirmed that simple kernel ridge regression matches or exceeds deep neural networks in-domain prediction, motivating the case for pre-training on large-scale, multiparadigm, data as a stronger inductive prior.

Tools: PyTorch, scikit-learn, HCP4AI HPC cluster; HCP-YA and CamCAN datasets.

Research Fellowship

Mar 2024 – Sep 2024

University of Turin

Supervisors: Giorgio Audrito, Gianluca Torta

- Extended bachelor's-thesis heuristics for distributed team formation to prevent over- and under-recruiting; manuscript in preparation.
- Developed distributed Aggregate Computing algorithms for graph centrality estimation (PageRank) on web crawls, validated on graphs up to ~300k nodes and currently scaling to ~40M nodes.

Tools: C++, MPI, Occam HPC cluster (UniTO).

Technical Skills

Programming Python, C++, C#

ML / DL PyTorch, scikit-learn

HPC SLURM, MPI; experience using HCP4AI, and Occam clusters

Other Git, L^AT_EX, Linux, Results visualization and presentation